Emerson & Cuming
a National Starch & Chemical Company

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Technical Data Sheet

STYCAST® 1265 A/B

Clear, Soft Gel, Epoxy Encapsulant

Key Feature:	Benefit:
• Long pot life	• Ease of use
• Soft gel	• Can be cut to repair or replace embedded components
• Low exotherm	• Results in low embedment stress

Product Description:

STYCAST 1265 A/B is a clear, two component epoxy gel. It is tough, has a hardness of 25 Shore A and a long pot life. The STYCAST 1265 A/B has excellent adhesion to a wide variety of substrates and can be cut easily for repairing and replacing components. Fresh material can be poured and then cured to form a perfect repair.

Applications:

The STYCAST 1265 A/B is designed to be used for embedding stress sensitive components or where inspections and repairs are desired. It may be used as a flexible adhesive at temperatures above 65 °C but flexibility will be decreased, a darkening of the color and a substantial decrease in electrical properties can be expected.

Instructions For Use:

Thoroughly read the information concerning health and safety contained in this bulletin before using. Observe all precautionary statements that appear on the product label and/or contained in individual Material Safety Data Sheets (MSDS).

To ensure the long term performance of the potted or encapsulated electrical / electronic assembly, complete cleaning of components and substrates should be performed to remove contamination such as dust, moisture, salt, and oils which can cause electrical failure, poor adhesion or corrosion in an embedded part.

Accurately weigh resin and hardener into a clean container in the recommended ratio. Weighing apparatus having an accuracy in proportion to the amounts being weighed should be used.

Blend components by hand, using a kneading motion, for 2-3 minutes. Scrape the bottom and sides of the mixing container frequently to produce a uniform mixture. If possible, power mix for an additional 2-3 minutes. Avoid high mixing speeds which could entrap excessive amounts of air or cause overheating of the mixture resulting in reduced working life.

To ensure a void-free embedment, vacuum deairing should be used to remove any entrapped air introduced during the mixing operation. Vacuum deair mixture at 1-5 mm mercury. The foam will rise several times the liquid height and then subside. Continue vacuum deairing until most of the bubbling has ceased. This usually requires 3-10 minutes.

Pour mixture into cavity or mold. Gentle warming of the mold or assembly reduces the viscosity. This improves the flow of the material into the unit having intricate shapes or tightly packed coils or components. Further vacuum deairing in the mold may be required for critical applications.

Properties of Material As Supplied:

Property	Test Method	Unit	Value - Part A	Value - Part B
Chemical Type			Epoxy	Amine
Appearance	Visual		Clear	Clear
Density	ASTM-D-792	g/cm^3	1.21	0.98
Brookfield Viscosity	ASTM-D-2393	Pa.s cP	10 10,000	0.23 230

Properties of Material As Mixed:

Property	Test Method	Unit	Value
Mix Ratio - Amount of Part B per 100 parts of Part A		By Weight	100
Working Life (100 g @ 25°C)	ERF 13-70	hours	>24
Density	ASTM-D-792	g/cm^3	1.08
Brookfield Viscosity	ASTM-D-2393	Pa.s cP	0.6 600

"Our service engineers are available to help purchasers obtain best results from our products, and recommendations are based on tests and information believed to be reliable. However, we have no control over the conditions under which our products are transported to, stored, handled, or used by purchasers and, in any event, all recommendations and sales are made on condition that we will not be held liable for any damages resulting from their use. No representative of ours has any authority to waive or change this provision. We also expect purchasers to use our products in accordance with the guiding principles of the Chemical Manufacturers Association's Responsible Care® program."

Cure Schedule:

Cure at any one of the recommended cure schedules. For large castings, use the lower temperature cures. Cured hardness is somewhat dependent on cure temperature. Lower temperatures generally result in a softer material.

Temperature	Cure Time
°C	Time
25	3 to 7 days
45	48 hours
65	16 hours
95	4 hours

Properties of Material After Application:

Property	Test Method	Unit	Value
Hardness	ASTM-D-2240	Shore A	25
Temperature Range of Use		°C	-40 to +65
Dielectric Constant @ 1 mHz	ASTM-D-150	--	3.0
Dissipation Factor @ 1 mHz	ASTM-D-150	-	0.02
Volume Resistivity @ 25°C	ASTM-D-257	Ohm-cm	10^{12}

Storage and Handling:

The shelf life of STYCAST 1265 A/B is 12 months at 25°C. For best results, store in original, tightly covered containers. Storage in cool, clean and dry areas is recommended. Usable shelf life may vary depending on method of application and storage temperature.

Health and Safety:

The STYCAST 1265 Part A like most epoxy compounds possesses the ability to cause skin and eye irritation upon contact. Certain individuals may also develop an allergic reaction after exposure (skin contact, inhalation of vapors, etc.) which may manifest itself in a number of ways including skin rashes and an itching sensation. Handling this product at elevated temperatures may also generate vapors irritating to the respiratory system.

The STYCAST 1265 Part B is classified as a corrosive material. Direct contact with unprotected eyes or skin can cause severe burns. Handling this product may also generate vapors irritating to the respiratory system. Components of this material may be absorbed through the skin.

Good industrial hygiene and safety practices must be used when handling this product. Proper eye protection and appropriate chemical resistant clothing must be worn to prevent contact and possible skin absorption. Consult the Material Safety Data Sheet (MSDS) for detailed recommendations on the use of engineering controls , personal protective equipment and first aid procedures.

This information is only a brief summary of the available safety and health data. Thoroughly review the MSDS for more complete information before using this product.

Attention Specification Writers:
The values contained herein are considered typical properties only and are not intended to be used as specification limits. For assistance in preparing specifications, please contact Emerson & Cuming Quality Assurance for further details.

■ Underfills Solder Alternatives C.O.B. Materials Film Adhesives Thermal Interfaces ■

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■ Encapsulants Coatings Adhesives Electrically Conductive Coatings and Adhesives ■

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